

Individual Homework 2 — Applied ML for Business

Networks Module | Individual | 66 points

Submission: Upload [studentid].md on Moodle (DLE). Answers must be typed in markdown format. Use the question numbering exactly as it appears here; matching is essential for grading.

On code: Run all code on your own machine. Submit only the outputs and your interpretations — not the code itself. Where a question asks you to compute something, report the result and then answer the interpretive question. No plots or images required in your submission file.

Individual submission. Spend adequate time with each question; the goal is understanding, not completion speed.

Data files (course GitHub, hw2/data/): - `cs_hiring_edges.csv` — directed hiring edges for CS departments (`from_dept`, `to_dept`, `count`) - `cs_hiring_nodes.csv` — CS department names and prestige ranks - `biz_hiring_edges.csv` — directed hiring edges for Business departments - `biz_hiring_nodes.csv` — Business department names and prestige ranks

Edge direction: an edge from A to B means department A placed one of its PhD graduates as faculty at department B.

Q1 — The Hiring Network [10 pts]

Academic departments hire their faculty from other departments. The network of these placements reveals the structure of academic prestige: where does talent flow, and which programs sit at the centre of that flow?

- (a) [2 pts] Report the number of nodes and edges in the CS hiring network and in the Business hiring network.
- (b) [3 pts] Compute network density for both networks. For a directed graph: $\text{density} = \text{edges} / n(n-1)$. Compare the two densities, and offer one substantive explanation for what you find.
- (c) [2 pts] What is the average in-degree in each network? What does a department's in-degree represent in this context?
- (d) [3 pts] Find the top 5 departments by in-degree in CS and in Business. Do the highest in-degree departments match the top prestige ranks in the nodes file? Describe what you observe in 2–3 sentences.

Q2 — Eigenvector Centrality [12 pts]

A department is important not just because it hires many people, but because it hires from other important departments. Eigenvector centrality captures this mutual-endorsement logic: it is the principal eigenvector of the adjacency matrix, satisfying $Ac = \lambda c$.

- (a) [2 pts] Implement power iteration on the CS adjacency matrix and report the top 5 departments by EC score.

- (b) [3 pts] Compute the Spearman rank correlation between your EC ranking and the prestige ranks in `cs_hiring_nodes.csv`. Report the correlation and interpret it in one sentence.
- (c) [3 pts] Repeat for the Business network. Report the top 5 EC departments and the Spearman correlation with prestige.
- (d) [4 pts] Compare the two Spearman correlations from Q2(b) and Q2(c). What structural property of a network explains the difference you find? Point to a specific number from Q1 that supports your answer.
-

Q3 — PageRank [14 pts]

PageRank extends eigenvector centrality to directed graphs. A department's score depends on the scores of departments that send it graduates — it measures the quality of a program's academic genealogy.

Use $\alpha = 0.85$ throughout.

- (a) [2 pts] Run PageRank on the CS hiring network using the forward edges. Report the top 5 departments by forward PageRank.
- (b) [3 pts] Reverse all edges and run PageRank again. Report the top 5 departments by reversed PageRank. In one sentence, state what reversed PageRank measures in the hiring context.
- (c) [3 pts] Compute the Spearman correlation between reversed PageRank and prestige for the CS network. How does this compare to the correlation you found using EC in Q2(b)?
- (d) [3 pts] Find DePaul University's rank under forward PageRank and under reversed PageRank. Report both ranks and the gap. Explain the structural reason: what does DePaul's position in the hiring network look like, and what does the gap say about how it fits into the CS academic system?
- (e) [3 pts] An administrator at a lower-ranked program argues: "We should publicise our PageRank score — it shows we're well-connected in the network." A colleague replies: "Your PageRank is high because elite programs send you graduates as a last resort, not because you're prestigious." What does this exchange reveal about using forward PageRank as a prestige signal?
-

Q4 — Betweenness Centrality and Community Detection [8 pts]

A UK gift wholesaler collected 12 months of online sales records. Each transaction lists the products bought together in one basket. Represent this as a product co-purchase network: products are nodes; two products are connected if they were frequently bought together. The network has 2,801 products and 530,267 co-purchase pairs. Running community detection yields four communities — Home Décor & Gifting (1,060 products), Christmas / Seasonal (678), Kitchen / Retrospect (577), Craft & Traditional (496) — with modularity $Q = 0.16$. The product with the highest betweenness centrality is WHITE HANGING HEART T-LIGHT HOLDER.

- (a) [2 pts] What does high betweenness centrality mean for a product in a co-purchase network? What kind of shopping behaviour would produce a product with unusually high BC?

(b) [3 pts] Is $Q = 0.16$ a strong community structure or a weak one? What does this specific value tell you about how distinctly the four product groups cluster relative to what you would expect by chance?

(c) [3 pts] A retail analyst wants to use community membership as a feature in a recommendation model. She trains on the first 9 months and evaluates on the last 3. Describe one concrete reason why a product's community label might not transfer cleanly from training to test, and what she should check before using community as a feature.

Q5 — Machine Learning on Networks [6 pts]

The U.S. Centers for Medicare and Medicaid Services (CMS) publishes provider-level payment data. Providers who share patients are connected in a network. The Office of Inspector General (OIG) excludes providers found guilty of fraud. An ML team builds a classifier to flag suspicious providers. They have two groups of features: - **Group A (structural)**: degree, PageRank, community membership - **Group B (label propagation)**: `neighbor_fraud_rate` — the fraction of a provider's network neighbours who are OIG-excluded

A model trained on Group A alone achieves $F1 = 0.37$. Adding `neighbor_fraud_rate` pushes F1 to 0.60.

(a) [3 pts] A colleague says: “Both groups use network structure — they’re the same kind of feature.” Explain clearly why they are different, and why the difference matters for model training and evaluation.

(b) [3 pts] The team considers replacing Group A features with a 1,536-dimensional embedding — generated by passing each provider's CMS billing description (specialty, procedure mix, reimbursement patterns) through a language model. What new signal would these embeddings capture that the structural features in Group A miss? What would they still fail to capture even after two rounds of graph message passing?

Q6 — Temperature and Exploration [8 pts]

In the lecture we connected the softmax function to random walks on a graph. From node i with neighbours $j \in N(i)$:

$$P(j \mid i) = \frac{e^{z_{ij}/T}}{\sum_{k \in N(i)} e^{z_{ik}/T}}$$

where z_{ij} is a similarity score between i and j , and $T > 0$ is a temperature parameter.

(a) [2 pts] A user node has three neighbours with similarity scores $z = [2.0, 1.5, 0.5]$. Compute $P(j \mid i)$ for $T = 0.5$ and for $T = 2.0$. Report both distributions (rounded to 2 decimal places).

(b) [3 pts] For $T = 2.0$, describe what a random walk over many steps looks like compared to $T = 0.5$. Will the walk eventually reach every node? Does your answer change if the graph has two disconnected components?

(c) [3 pts] A product manager says: “We should always use low temperature — users get what they want.” A data scientist replies: “Low temperature may hurt long-run engagement.” What specifically does higher temperature enable that repeatedly recommending the highest-scoring item cannot? Give one concrete example from a platform context.

Q7 — Open Problem Pitch [8 pts]

Identify a real Indian business or societal problem where a **graph-based machine learning model** would outperform a standard tabular model on a prediction task. Your answer should be 180–250 words and must cover:

1. **The prediction task.** What outcome are you predicting?
2. **The network.** What are the nodes and edges? What relationship do the edges encode?
3. **The data.** Where do the node features come from? Where do the labels come from? Name actual sources — a government portal, exchange filing, regulatory database, public API. The data need not be collected, but it must exist and be accessible. Do not invent it.
4. **Why graph-based ML.** What does the graph structure add that a tabular model trained on node features alone would miss? Be specific about what signal travels through the edges.

Grading. Marks go to: (a) how sharp and non-obvious the problem is, (b) how concrete and realistic the data sources are for both features and labels, and (c) how clearly you argue why graph structure carries predictive signal a tabular model cannot recover. This is the only question in this assignment the instructor will read personally — not an AI grader.

Total: 66 points